

An Approximate Capacity Bound for Sparse Combinatorial Key-Value Memory: Full Derivation and Validation

Supplementary note to “Combinatorial Coding, Key-Value Recall, and Emergent
Constraint Satisfaction in a Hippocampal-Inspired Spiking Network”

September 17, 2026

Abstract

This note gives the complete derivation, with every intermediate step made explicit, of the approximate capacity bound used in the main paper (Section 2.3): an expression for the largest number of items M a sparse K -of- H combinatorial key-value memory can store while remaining reliably retrievable under a given level of query corruption. The bound is structurally the same as modern Hopfield networks’ proven exponential capacity [1], but is derived here for a discrete, sparse code via elementary combinatorics and extreme value theory rather than for dense continuous patterns via an energy-function argument. We include a full, self-contained proof of the one non-elementary step (the expected maximum of many correlated-in-effect-but-treated-as-independent Gaussian similarities), validate the final bound against simulation, and discuss its relationship to the wider family of capacity theorems it belongs to. We then go further than that first-moment approximation: Section 8 derives an exact, non-asymptotic failure probability directly from the hypergeometric distribution, and a corresponding rigorous sufficient bound on capacity at any chosen target reliability δ , correcting a subtle correlation error that a natural-looking marginalization over query corruption would otherwise introduce. This exact bound is, as expected, considerably more conservative than the first-moment approximation, and is shown to be tight in the well-separated regime and honestly, quantifiably loose only as the system approaches its collapse transition.

1 Setup

We store M items. Item m ’s key is a binary vector $h_m \in \{0, 1\}^H$ with exactly K active (value-1) coordinates – a K -of- H sparse code. Throughout this note we model the M stored keys as independent, uniformly random K -subsets of $[H] = \{1, \dots, H\}$: an idealization relative to the true architecture, where keys are W_{SH} -projections of mutually correlated combinatorial structure (solved Sudoku grids), addressed explicitly in Section 9.

A query is formed from the true key h_{true} by flipping a fraction $\eta \in [0, 1]$ of all H coordinates, chosen uniformly at random without replacement – the noise model used throughout the main paper. Retrieval compares the query against all M stored keys via similarity (shared active coordinates) and should return the index of h_{true} . Retrieval *fails* when some other stored key’s similarity to the query exceeds h_{true} ’s.

We ask: for fixed H , K , and corruption level η , how large can M be before this kind of failure becomes likely?

2 Step 1: Similarity between two independent random keys

Lemma 1 (Hypergeometric overlap). *Let $A, B \subset [H]$ be independent uniformly random subsets with $|A| = a$, $|B| = b$. Then $|A \cap B|$ is a hypergeometric random variable with*

$$\mathbb{E}[|A \cap B|] = \frac{ab}{H}, \quad \text{Var}(|A \cap B|) = a \cdot \frac{b}{H} \left(1 - \frac{b}{H}\right) \cdot \frac{H - a}{H - 1}.$$

Proof sketch. Fix A and treat B 's b elements as drawn sequentially without replacement from $[H]$. Let $X_i = \mathbb{1}[\text{the } i\text{-th draw lands in } A]$, so $|A \cap B| = \sum_{i=1}^b X_i$. Each X_i is Bernoulli(a/H) by symmetry, giving the mean by linearity of expectation. For the variance, $\text{Cov}(X_i, X_j) < 0$ for $i \neq j$ because sampling without replacement makes draws negatively associated (having drawn a marked element makes drawing another marked element on a later draw slightly less likely); summing the b variances and $\binom{b}{2}$ covariances gives the stated finite-population-corrected formula. (Full derivation: Feller, *An Introduction to Probability Theory*, Vol. 1, Ch. II.) $\square$

Applying Lemma 1 with $a = b = K$ gives the similarity between two *uncorrupted* stored keys: mean K^2/H , the number quoted throughout the main paper (e.g. $K^2/H = 22.5$ for the default $H = 1000$, $K = 150$).

3 Step 2: Statistics of a corrupted query

Flipping a uniformly random fraction η of all H coordinates is itself an instance of Lemma 1: the ηH flipped positions are a random subset of $[H]$, and we ask how many of h_{true} 's K active positions they contain. By the lemma (with $a = K$, $b = \eta H$), the expected number of active positions flipped *off* is ηK , so

$$S_{\text{true}}(\eta) = \mathbb{E}[\text{similarity of query to } h_{\text{true}}] = K(1 - \eta), \quad (1)$$

exactly (no further approximation needed: this is a direct expectation, not a bound). The expected number of originally-*inactive* positions flipped *on* is, symmetrically, $\eta(H - K)$, so the corrupted query's own total active-coordinate count is, in expectation,

$$K_q(\eta) = K(1 - \eta) + \eta(H - K) = K(1 - 2\eta) + \eta H. \quad (2)$$

The query's similarity to any *other*, independent stored key is now Lemma 1 applied with $a = K_q(\eta)$, $b = K$:

$$\mu_{\text{other}}(\eta) = \frac{K_q(\eta) K}{H}, \quad \sigma_{\text{other}}^2(\eta) = K_q(\eta) \frac{K}{H} \left(1 - \frac{K}{H}\right) \frac{H - K_q(\eta)}{H - 1}. \quad (3)$$

4 Step 3: Gaussian approximation

The hypergeometric distribution of S_{other} (similarity to one specific other key) is exact but discrete and unwieldy for the extreme-value step that follows. We approximate

$$S_{\text{other}} \approx \mathcal{N}(\mu_{\text{other}}(\eta), \sigma_{\text{other}}^2(\eta)),$$

justified for $H \gg 1$ by the central limit theorem for sampling without replacement from a finite population [2, 3]: sums arising from simple random sampling without replacement are asymptotically normal as the population size grows, with exactly the finite-population-corrected variance already appearing in Lemma 1. This is the least exact step in the derivation, and its error is largest precisely when σ_{other} is small relative to the mean – i.e. at low noise, where the discreteness of the underlying hypergeometric distribution matters most. Section 7 quantifies this error directly rather than leaving it as a caveat.

5 Step 4: The maximum of many competitors

Retrieval does not fail when the *typical* competing key looks similar to the query – it fails when the *single most similar* of the $M - 1$ competitors exceeds $S_{\text{true}}(\eta)$. This is an extreme-value criterion, and it is the step that most distinguishes this derivation from a naive signal-to-noise argument.

Lemma 2 (Expected maximum of i.i.d. Gaussians). *Let $Z_1, \dots, Z_n$ be i.i.d. $\mathcal{N}(0, 1)$. Then*

$$\mathbb{E} \left[\max_{1 \leq i \leq n} Z_i \right] \leq \sqrt{2 \ln n} + o(1) \quad \text{as } n \rightarrow \infty,$$

and this bound is asymptotically tight: $\mathbb{E}[\max_i Z_i] = \sqrt{2 \ln n} (1 + o(1))$.

Proof of the upper bound. For any $t \geq 0$, the standard Gaussian tail (Chernoff) bound gives $\Pr(Z_1 > t) \leq e^{-t^2/2}$: with $s = t$, $\Pr(Z_1 > t) = \Pr(e^{sZ_1} > e^{st}) \leq \mathbb{E}[e^{sZ_1}]e^{-st} = e^{s^2/2-st}$, minimized at $s = t$. A union bound over the n variables gives $\Pr(\max_i Z_i > t) \leq \min(1, ne^{-t^2/2})$. Writing the expectation of a nonnegative random variable as an integral of its tail,

$$\mathbb{E} \left[\max_i Z_i \right] = \int_0^\infty \Pr \left(\max_i Z_i > t \right) dt \leq \int_0^{t^*} 1 dt + \int_{t^*}^\infty ne^{-t^2/2} dt,$$

where $t^* = \sqrt{2 \ln n}$ is chosen so the two bounds on the integrand meet ($ne^{-(t^*)^2/2} = 1$). The first integral is exactly $t^* = \sqrt{2 \ln n}$. For the second, the standard Gaussian tail estimate $\int_a^\infty e^{-t^2/2} dt \leq \frac{1}{a} e^{-a^2/2}$ (for $a > 0$, obtained by writing $e^{-t^2/2} \leq \frac{t}{a} e^{-t^2/2}$ on $t \geq a$ and integrating exactly) gives

$$\int_{t^*}^\infty ne^{-t^2/2} dt \leq \frac{n}{t^*} e^{-(t^*)^2/2} = \frac{1}{\sqrt{2 \ln n}} \xrightarrow{n \rightarrow \infty} 0.$$

Summing the two pieces gives $\mathbb{E}[\max_i Z_i] \leq \sqrt{2 \ln n} + o(1)$. The matching lower bound requires a second-moment (Paley–Zygmund-type) argument and is omitted here; see Boucheron, Lugosi & Massart, *Concentration Inequalities* [4], Ch. 2, for the complete two-sided result. $\square$

Rescaling Lemma 2 to $\mathcal{N}(\mu_{\text{other}}, \sigma_{\text{other}}^2)$ variables and setting $n = M - 1$ (the number of competing stored keys) gives the working approximation used in the main paper:

$$\mathbb{E} \left[\max_{i=1}^{M-1} S_{\text{other},i} \right] \approx \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}. \quad (4)$$

6 Step 5: Solving for capacity

Retrieval is (approximately, at the level of expectations rather than a proper tail bound on failure probability) reliable when $S_{\text{true}}(\eta)$ exceeds the expected maximum competitor, Equation 4. Define the normalized separation margin

$$z(\eta) = \frac{S_{\text{true}}(\eta) - \mu_{\text{other}}(\eta)}{\sigma_{\text{other}}(\eta)}. \quad (5)$$

Setting $S_{\text{true}}(\eta) = \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}$ and solving for M :

$$z(\eta) = \sqrt{2 \ln(M-1)} \iff M-1 = \exp\left(\frac{z(\eta)^2}{2}\right) \iff \boxed{M_{\text{max}}(\eta) \approx 1 + \exp\left(\frac{z(\eta)^2}{2}\right)}. \quad (6)$$

Proposition 1 (Structural form). *Equation 6 states that capacity is exponential in the square of a normalized signal-to-noise-like quantity. This is the same structural form as modern Hopfield networks’ proven capacity $C \cong 2^{d/2}$ [1], though that bound is derived via a Lyapunov energy-function argument (the Concave–Convex Procedure applied to a log-sum-exp energy) over dense continuous patterns, not via hypergeometric statistics over a sparse discrete code.*

7 Validation against simulation

Equations 1–6 were implemented exactly as derived (`capacity_bound_derivation.py`, no free parameters fit to data) and compared against the empirically measured collapse thresholds reported in the main paper.

	$M=50$	$M=1000$	$M=10,000$	$K=10$	$K=150$
Predicted (this note, $H=1000$)	43.9%	41.9%	40.6%	24.5%	41.9%
Observed (simulation)	45%	45%	40%	25%	40–45%

Agreement is within 1–5 percentage points across a 200-fold range of M and an order of magnitude in K , including an almost exact match at $K = 10$. A second, independent check: at $H=1000$, $K=150$, $M=200$, the derivation predicts the ratio $S_{\text{true}}(\eta)/\mathbb{E}[\max_i S_{\text{other},i}]$ at $\eta \in \{0, 0.20, 0.30\}$ as 4.21, 1.99, 1.47 respectively, against independently measured ratios of 3.39, 1.92, 1.45 (obtained from a completely separate experiment, the separation-operator ablation, not used to fit any parameter of this derivation). The two noisy points match to within 4% and 1.4%; the noise-free point is over-predicted by about 24%, consistent with Section 4’s observation that the Gaussian approximation is weakest exactly there.

8 An exact hypergeometric tail bound

Equation 6 matches only the *expected* maximum competitor similarity, not a probabilistic guarantee on retrieval failure – a gap noted below in Section 9. This section closes it with an exact, non-asymptotic tail bound on the failure probability itself, derived directly from the hypergeometric distribution rather than its Gaussian approximation, and valid at any finite M .

8.1 A correlation a naive marginalization misses

Let R denote the (random) number of h_{true} ’s K active positions that land among the ηH flipped query positions. By Lemma 1 (with $a = K$, $b = \eta H$), R is exactly hypergeometric:

$$R \sim \text{Hypergeometric}(H, K, \eta H), \quad \Pr(R = r) = \frac{\binom{K}{r} \binom{H-K}{\eta H - r}}{\binom{H}{\eta H}}.$$

Conditional on $R = r$, both the query’s active-coordinate count and its similarity to h_{true} are *deterministic*, not merely expected:

$$k(r) = K + \eta H - 2r, \quad s(r) = K - r,$$

the realized versions of Equations 2 and 1, which recover $k(\eta K) = K_q(\eta)$ and $s(\eta K) = S_{\text{true}}(\eta)$ only in expectation. Crucially, $k(r)$ and $s(r)$ are perfectly correlated through the same draw of r :

a query realization with more of its true bits flipped off (large r) simultaneously has lower true-similarity *and* a different active count. A natural-looking but invalid shortcut is to hold $s(r)$ fixed at its unconditional expectation $S_{\text{true}}(\eta) = K(1 - \eta)$ while still letting $k(r)$ vary with r inside the union bound over competitors: since $k(r)$ and $s(r)$ are deterministic functions of the very same random variable, fixing one while randomizing the other breaks their correlation and is not a valid marginalization of any well-defined random process. Numerically, this shortcut understates the true failure probability by roughly four to five orders of magnitude for $\eta \in [0.30, 0.40]$, converging to the correct value only as η approaches the collapse region (where r 's distribution concentrates and the distinction between fixing $s(r)$ at r or at $\mathbb{E}[r]$ stops mattering).

8.2 Exact per-realization failure probability

Fix one specific other stored key h_j ($j \neq \text{true}$), itself an independent random K -subset of $[H]$. Conditional on $R = r$, the query is a specific, realized $k(r)$ -subset, and its overlap with h_j is exactly hypergeometric by Lemma 1 (with $a = k(r)$, $b = K$). Define the exact tail probability

$$p_k(s) := \Pr [\text{Hypergeometric}(H, k, K) \geq s] = \sum_{i=s}^{\min(k, K)} \frac{\binom{k}{i} \binom{H-k}{K-i}}{\binom{H}{K}}.$$

(We use $\geq$ rather than strict $>$: an exact tie is conservatively counted as a failure, which can only inflate the resulting probability and keeps Proposition 2 below a genuine upper bound regardless of how ties are broken at retrieval time.) Conditional on $R = r$, the probability that this one specific competitor's similarity meets or exceeds h_{true} 's is exactly $p_{k(r)}(s(r))$, and with $M - 1$ independent competitors the conditional failure probability is exactly

$$\Pr[\text{fail} \mid R = r] = 1 - [1 - p_{k(r)}(s(r))]^{M-1}.$$

Averaging over R gives the exact unconditional failure probability – no Gaussian approximation, no expectation-of-a-maximum heuristic, and no fixing of $s(r)$ at its mean:

$$P_{\text{fail}}(M, \eta) = \sum_{r=0}^K \Pr(R = r) \cdot \left\{ 1 - [1 - p_{k(r)}(s(r))]^{M-1} \right\}. \quad (7)$$

Equation 7 is exact given the independence assumption of Section 9; it involves only a finite sum over $r \in \{0, \dots, K\}$ and is cheap to evaluate for any H, K, η, M .

A small-tail approximation and a characteristic scale. Away from the collapse transition, $p_{k(r)}(s(r))$ is typically small, and

$$[1 - p_{k(r)}(s(r))]^{M-1} \approx \exp[-(M-1)p_{k(r)}(s(r))],$$

so, conditional on $R = r$, retrieval probability decays approximately exponentially in M with a characteristic scale

$$M_r^* \sim \frac{1}{p_{k(r)}(s(r))}.$$

Intuitively: a dangerous competitor is individually unlikely at any single comparison, but there are $M - 1$ near-independent opportunities for one to occur, and their cumulative probability grows with memory load – the same “many rare independent chances” structure behind a Poisson approximation, applied here to the union of $M - 1$ competitor comparisons. This is an approximation used only for intuition; Equation 7 itself requires no such approximation.

8.3 No finite-size singularity: monotonicity and convexity in M

Equation 7 is not just numerically well-behaved (Section 8.5 below) – it is provably free of any finite- M singularity of the kind a genuine attractor-collapse memory cliff would produce. This is a structural property of the exact formula itself, independent of the union bound derived in the next section, and is worth stating as a theorem in its own right.

Theorem 1 (No finite-size singularity). *Fix H , K , and η , and suppose $0 < p_{k(r)}(s(r)) < 1$ for every r with $\Pr(R = r) > 0$ (against any realized corruption level, a single competitor’s failure is neither impossible nor certain). Extend Equation 7 to continuous $M \geq 1$ via*

$$P_{\text{ret}}(M, \eta) := 1 - P_{\text{fail}}(M, \eta) = \sum_{r=0}^K \Pr(R = r) [1 - p_{k(r)}(s(r))]^{M-1}.$$

Then $P_{\text{ret}}(M, \eta)$ is strictly decreasing and strictly convex in M :

$$\frac{\partial P_{\text{ret}}}{\partial M} < 0, \quad \frac{\partial^2 P_{\text{ret}}}{\partial M^2} > 0.$$

Proof. Write $q_r := 1 - p_{k(r)}(s(r)) \in (0, 1)$ for each r with $\Pr(R = r) > 0$. Treating M as a continuous variable, $q_r^{M-1} = \exp[(M-1) \ln q_r]$, so

$$\frac{\partial}{\partial M} q_r^{M-1} = q_r^{M-1} \ln q_r, \quad \frac{\partial^2}{\partial M^2} q_r^{M-1} = q_r^{M-1} (\ln q_r)^2.$$

Since $q_r \in (0, 1)$, $\ln q_r < 0$ and $q_r^{M-1} > 0$, so the first derivative is strictly negative and the second, a square of a nonzero real number times a positive factor, is strictly positive. $P_{\text{ret}}(M, \eta)$ is a finite sum of these terms weighted by $\Pr(R = r) \geq 0$ summing to 1; differentiation is linear and commutes with a finite sum, so

$$\frac{\partial P_{\text{ret}}}{\partial M} = \sum_r \Pr(R = r) q_r^{M-1} \ln q_r < 0, \quad \frac{\partial^2 P_{\text{ret}}}{\partial M^2} = \sum_r \Pr(R = r) q_r^{M-1} (\ln q_r)^2 > 0,$$

each a weighted average, with at least one strictly positive weight, of terms that are all strictly negative (respectively all strictly positive). $\square$

Interpretation. Theorem 1 rules out, at the level of the exact model, any finite- M point at which retrieval probability could drop discontinuously or bend toward an accelerating collapse: P_{ret} can only *decelerate* as M grows (convexity), on top of always decreasing (monotonicity) – graceful, non-catastrophic degradation, now proven rather than only observed in simulation (Section 8.5). This is a property specific to the *explicit-key* retrieval mechanism analyzed throughout this note: failure accumulates from more chances for some one unrelated stored key to accidentally resemble the query, not from the appearance or disappearance of a dynamical attractor – the latter being the mechanism behind a genuine finite-size memory cliff in attractor-based (Hopfield-type) associative memories, and a distinction this note does not claim to resolve for that broader model class (Section 9).

8.4 A rigorous, closed-form sufficient bound

Equation 7 is exact but implicit: it gives P_{fail} as a function of M , not M as a function of a target failure probability δ . We invert it via the union bound, applied *per realization of r* before averaging – the order matters, and doing it in this order is what keeps the resulting bound valid.

By Bernoulli's inequality, $1 - (1 - p)^n \leq np$ for any $p \in [0, 1]$ and integer $n \geq 0$. Applied pointwise, for every r ,

$$1 - [1 - p_{k(r)}(s(r))]^{M-1} \leq (M-1)p_{k(r)}(s(r)).$$

This is a deterministic inequality between two functions of r ; taking $\mathbb{E}_R[\cdot]$ of both sides preserves it, since expectation is monotone:

$$P_{\text{fail}}(M, \eta) \leq (M-1) \cdot \underbrace{\mathbb{E}_R[p_{k(R)}(s(R))]}_{=: \bar{P}(\eta)}. \quad (8)$$

Proposition 2 (Corrected sufficient capacity bound). *For any target failure probability $\delta \in (0, 1)$, storing*

$$M \leq M_{\max}(\eta, \delta) := 1 + \left\lfloor \frac{\delta}{\bar{P}(\eta)} \right\rfloor \quad (9)$$

items guarantees $P_{\text{fail}}(M, \eta) \leq \delta$ exactly, subject only to the independence assumption of Section 9 – not to any Gaussian or large- M asymptotic approximation.

Proof. Immediate from Equation 8: $(M-1)\bar{P}(\eta) \leq \delta$ whenever $M-1 \leq \delta/\bar{P}(\eta)$, i.e. $M \leq 1 + \lfloor \delta/\bar{P}(\eta) \rfloor$. $\square$

Equation 9 has the same union-bound shape as a natural first attempt at this formula, $1 + \lfloor \delta/p_{K_q(\eta)}(S_{\text{true}}(\eta)) \rfloor$, but with p correctly taken to be $\bar{P}(\eta) = \mathbb{E}_R[p_{k(R)}(s(R))]$ – the per-realization tail probability *properly averaged* over R – rather than a single tail probability evaluated at the mismatched pair $(K_q(\eta), S_{\text{true}}(\eta))$, which (Section 8.1) treats the numerator and denominator of the same random draw as independently settable.

8.5 Numerical validation

Table 1 reports $\bar{P}(\eta)$ and the resulting $M_{\max}(\eta, \delta)$ at $H = 1000$, $K = 150$ for two target failure probabilities, alongside $M_{\max}(\eta)$ from Equation 6 (Section 6) for comparison. All quantities were computed with `scipy.stats.hypergeom` and, where probabilities are extremely small, the numerically stable form $1 - (1 - p)^n = -\text{expm1}(n \cdot \log1p(-p))$ to avoid catastrophic cancellation.

η	$\bar{P}(\eta)$	$M_{\max}(\eta, \delta=0.01)$	$M_{\max}(\eta, \delta=0.05)$	$M_{\max}(\eta)$ [Eq. 6]
0.20	2.41×10^{-35}	4.15×10^{32}	2.07×10^{33}	2.29×10^{48}
0.30	8.26×10^{-15}	1.21×10^{12}	6.05×10^{12}	1.61×10^{19}
0.35	9.59×10^{-9}	1.04×10^6	5.21×10^6	2.61×10^{10}
0.40	1.22×10^{-4}	83.0	411	3.27×10^4
0.42	1.81×10^{-3}	6.0	28.0	739
0.45	3.73×10^{-2}	1.0	2.0	13.9

Table 1: Corrected exact sufficient bound versus the expected-maximum approximation (both at $H = 1000$, $K = 150$).

Two things stand out. First, the exact tail-based bound is dramatically more conservative than the expected-maximum approximation – by about seven orders of magnitude at $\eta = 0.30$ (1.2×10^{12} vs. 1.6×10^{19}), narrowing to about a factor of 120 at $\eta = 0.42$. This gap is expected, not a contradiction between the two derivations: Equation 6 only requires the true match to beat

the *expected value* of the maximum competitor, a much weaker requirement than controlling the actual *probability* of failure to a small δ via a union bound over $M - 1$ competitors. The two bounds answer different questions – “typical-case margin” versus “guaranteed failure rate” – and are best read together: Equation 6 explains *why* capacity is exponential in the squared margin (the mechanism), while Equation 9 gives an actually-defensible number at a chosen reliability level (the guarantee). Section 9’s prediction that a proper tail bound “would very likely give a somewhat smaller $M_{\max}$ ” is confirmed, considerably more starkly than that hedge anticipated.

Second, we check how loose the sufficient bound (Equation 9) is relative to the exact failure probability it bounds (Equation 7), by comparing $(M - 1)\bar{P}(\eta)$ against $P_{\text{fail}}(M, \eta)$ directly (Table 2).

η	M	$P_{\text{fail}}(M, \eta)$ [exact, Eq. 7]	$(M - 1)\bar{P}(\eta)$ [Eq. 8]	ratio
0.30	200	1.644×10^{-12}	1.644×10^{-12}	1.00
0.30	1000	8.252×10^{-12}	8.252×10^{-12}	1.00
0.30	4000	3.303×10^{-11}	3.303×10^{-11}	1.00
0.40	200	1.82×10^{-2}	2.42×10^{-2}	1.33
0.40	1000	5.98×10^{-2}	1.22×10^{-1}	2.03
0.40	4000	1.35×10^{-1}	4.87×10^{-1}	3.59
0.42	200	1.59×10^{-1}	3.61×10^{-1}	2.27
0.42	1000	3.46×10^{-1}	1.81×10^0	5.23
0.42	4000	5.38×10^{-1}	7.25×10^0	13.46

Table 2: Tightness of the sufficient bound against the exact failure probability.

The bound is essentially tight (ratio ≈ 1.00) at $\eta = 0.30$, and increasingly loose – by design, as any union bound must be once per-competitor probabilities are no longer small – as η approaches the collapse transition and M grows (up to a factor of 13.5 at $\eta = 0.42$, $M = 4000$). This is the expected and honest behavior of a union bound: tightest exactly in the well-separated regime where a capacity claim is most meaningful, and openly announcing its own growing conservatism as the system approaches saturation rather than failing silently.

9 Limitations

Two assumptions are false by construction and are stated as such rather than absorbed silently.

Independence. Real stored keys are W_{SH} -projections of genuinely correlated combinatorial objects (solved Sudoku grids share row/column/box structure), not independent uniform random subsets. The observed collapse thresholds sit consistently at or slightly above the predicted ones (Section 7); a plausible explanation is that this real correlation makes the worst-case competing key marginally less extreme than the independent model assumes, though this has not been tested directly.

The extreme-value step is a first-moment approximation, not a tail bound – now resolved. Equation 4 matches the *expected* maximum, not a high-probability bound on it. Section 8 closes this gap with an exact, non-asymptotic alternative (Equation 7) and a corresponding rigorous sufficient bound (Proposition 2), derived directly from the hypergeometric distribution rather than the Gaussian approximation or the expected-maximum heuristic. As anticipated, the rigorous bound gives a substantially smaller $M_{\max}$ at any fixed target failure probability – by several orders of magnitude away from the collapse region (Table 1) – and is itself provably tight in that same well-separated regime, growing conservatively (rather than silently failing) only as η approaches

collapse (Table 2). Equation 6 should now be read as characterizing the right *mechanism* – exponential capacity in the squared separation margin, hence a sharp rather than gradual collapse – while Equation 9 is the bound to use for an actual reliability guarantee at a chosen δ . Both bounds still share the independence assumption below, which remains open.

10 Outlook: an asymptotic large-deviation phase boundary

Theorem 1 rules out a finite- M singularity at fixed H . It does not rule out an asymptotically sharp threshold if H and M are taken to infinity together in a suitable exponential scaling regime – the same sense in which Shannon’s channel coding theorem [5] produces a sharp rate/capacity boundary only in the block-length limit, never at any finite block length. We record the expected shape of this limit here as motivation for future work, not as a proven result: the rate function below is not derived from the hypergeometric tail, and the statement that follows is conditional on its existence in the assumed form.

Let $\rho = K/H$ and suppose the per-competitor tail probability admits a large-deviation form

$$p_H(\eta, \rho) \stackrel{H \rightarrow \infty}{\sim} \exp[-H I(\eta, \rho)] \quad (10)$$

for some rate function $I(\eta, \rho) \geq 0$ (large-deviation forms of this kind are standard for tail probabilities of hypergeometric-type sampling, by analogy with Sanov’s and Cramér’s theorems for sampling without replacement, but $I(\eta, \rho)$ is not derived here). If the number of stored memories scales exponentially with H , $M = \exp(\alpha H)$, then substituting Equation 10 into $P_{\text{ret}} \approx [1 - p_H(\eta, \rho)]^{M-1} \approx \exp[-(M-1)p_H(\eta, \rho)]$ (the small-tail approximation above) gives, to leading order,

$$P_{\text{ret}} \approx \exp\left\{-\exp[H(\alpha - I(\eta, \rho))]\right\}, \quad (11)$$

a sharp threshold at

$$\alpha_c(\eta, \rho) = I(\eta, \rho) : \quad P_{\text{ret}} \xrightarrow{H \rightarrow \infty} \begin{cases} 1 & \alpha < I(\eta, \rho) \quad (\text{reliable retrieval}) \\ 0 & \alpha > I(\eta, \rho) \quad (\text{competitors overwhelm retrieval}) \end{cases} \quad (12)$$

with critical scaling at $\alpha = I(\eta, \rho)$.

This is not a contradiction of Theorem 1: a finite- H curve that is provably smooth and convex at every finite size can still become arbitrarily sharp in a singular joint limit. The correct summary is *no finite-size attractor-collapse cliff*, not *no asymptotic phase boundary under any scaling*. Deriving $I(\eta, \rho)$ explicitly from the hypergeometric tail – turning Equation 12’s qualitative picture into an explicit phase boundary in (η, ρ, α) space – would connect this note’s finite-size theory directly to the large-deviation/statistical-mechanics literature cited below, and is left to future work.

11 Relation to the wider literature

This derivation belongs to a family of results sharing one underlying question: does a correct answer stand out against the *extreme* value among exponentially or combinatorially many near-random alternatives? The same skeleton, worked out through different technical routes, underlies Shannon’s channel coding theorem [5], the Random Energy Model in spin-glass theory [6], Gardner’s perceptron capacity [7], and the classical Amit–Gutfreund–Sompolinsky analysis of Hopfield network capacity [8]. Kanerva’s original capacity analysis for sparse distributed memory [10] is the closest existing derivation in spirit to this one – also a sparse binary code, also analyzed via combinatorial (there, Hamming-ball volume) arguments – though via a different geometric route than the hypergeometric-overlap statistics used here.

12 Appendix: An independent derivation of MESH’s heteroassociative theorems

Section 4.5 of the main paper ties an empirically measured capacity ceiling to two specific published results, MESH’s Theorems 5.1 and 5.3 [11], governing the two directions of pseudoinverse-learned heteroassociation between the scaffold’s hidden layer and an arbitrary feature (content) layer. We derive both independently here, from elementary linear algebra, rather than only citing them – and show they are the same fact, M^+M acting as either the identity or a projector depending on which side of a rank comparison M sits, applied to the two different matrices (F , then H) that appear in the two heteroassociative directions.

Setup. Let $H \in \mathbb{R}^{N_H \times N_{\text{patts}}}$ have columns h^μ , $\mu = 1, \dots, N_{\text{patts}}$, the scaffold’s predefined hidden-layer states, and let $F \in \mathbb{R}^{N_F \times N_{\text{patts}}}$ have columns f^μ , the arbitrary content patterns heteroassociated with them. Heteroassociative pseudoinverse learning sets $W_{HF} = HF^+$ (feature→hidden) and $W_{FH} = FH^+$ (hidden→feature).

Lemma 3 (The pseudo-inverse “hat” matrix is the identity or a projector). *Let $M \in \mathbb{R}^{a \times b}$.*

- (a) *If M has full column rank ($\text{rank}(M) = b \leq a$), then $M^+M = I_b$, exactly.*
- (b) *If M has full row rank ($\text{rank}(M) = a \leq b$), then $P := M^+M$ is the orthogonal projector onto the row space of M , with $\text{rank}(P) = a$; in particular $P = I_b$ only if $a = b$, and otherwise $P \neq I_b$.*

Proof. (a) Full column rank gives the left pseudo-inverse $M^+ = (M^\top M)^{-1}M^\top$ ($M^\top M$ invertible since it is $b \times b$ of rank b), so $M^+M = (M^\top M)^{-1}M^\top M = I_b$. (b) Full row rank gives the right pseudo-inverse $M^+ = M^\top(MM^\top)^{-1}$ ($MM^\top$ invertible since it is $a \times a$ of rank a), so $P := M^+M = M^\top(MM^\top)^{-1}M$ is symmetric ($P^\top = M^\top((MM^\top)^{-1})^\top M = P$, since $MM^\top$ is symmetric) and idempotent ($P^2 = M^\top(MM^\top)^{-1}(MM^\top)(MM^\top)^{-1}M = P$), hence an orthogonal projector, with $\text{rank}(P) = \text{rank}(M) = a$. If $a < b$, P is a rank- a projector on $\mathbb{R}^b$ and cannot equal I_b (rank b). $\square$

12.1 Theorem 5.1, re-derived: exact recovery when $N_{\text{patts}} \leq N_F$

MESH’s Theorem 5.1 states that if F is full rank, an input of clean feature vectors perfectly reconstructs the hidden layer states through pseudoinverse heteroassociation, provided $N_{\text{patts}} \leq N_F$ – exactly the regime in which F (an $N_F \times N_{\text{patts}}$ matrix) can have full *column* rank N_{patts} .

Proposition 3 (MESH’s Theorem 5.1, re-derived). *If F has full column rank N_{patts} (possible since $N_{\text{patts}} \leq N_F$), then presenting the clean feature vector f^μ recovers the hidden state exactly: $\hat{h}^\mu := W_{HF}f^\mu = h^\mu$, for every μ .*

Proof. Writing $f^\mu = Fe_\mu$, Lemma 3(a) applied to F gives $F^+F = I_{N_{\text{patts}}}$, so

$$\hat{h}^\mu = W_{HF}f^\mu = HF^+Fe_\mu = H I_{N_{\text{patts}}} e_\mu = He_\mu = h^\mu.$$

$\square$

Unlike Theorem 5.3 below, this is an exact deterministic identity, not an expectation: it requires no distributional assumption on F or H at all beyond F ’s rank, and holds for every stored pattern individually, not merely on average. We verified it numerically (random Gaussian F and H) across four $(N_F, N_{\text{patts}}, N_H)$ triples spanning an order of magnitude in each dimension; the maximum reconstruction error was at floating-point precision ($\leq 2 \times 10^{-12}$) in every case, exactly as the exact identity predicts.

12.2 Theorem 5.3, re-derived: partial recovery when $N_{\text{patts}} > N_H$

The complementary regime reverses which matrix supplies the rank condition and which supplies the randomness. Assume: (i) $N_{\text{patts}} > N_H$ and H has full row rank N_H (guaranteed for the scaffold's own predefined states by construction [11], independent of F); (ii) the columns f^μ are independent, zero-mean, and each satisfies $(f^\mu)^\top f^\mu = N_F$ exactly (e.g. i.i.d. random $\{-1, 1\}$ patterns), and are independent of H . Presenting the exact hidden state h^μ recovers $\hat{f}^\mu = W_{FH} h^\mu = FH^+ h^\mu$.

Proposition 4 (MESH's Theorem 5.3, re-derived). *Under assumptions (i)–(ii),*

$$\frac{1}{N_{\text{patts}}} \sum_{\mu=1}^{N_{\text{patts}}} \mathbb{E} \left[\frac{(f^\mu)^\top \hat{f}^\mu}{N_F} \right] = \frac{N_H}{N_{\text{patts}}}.$$

Proof. Writing $h^\mu = H e_\mu$, Lemma 3(b) applied to H gives $\hat{f}^\mu = FH^+ H e_\mu = F c^\mu$ where $c^\mu := P e_\mu$ and $P := H^+ H$ is the rank- N_H orthogonal projector onto H 's row space. The vector c^μ is column μ of P , so its own μ -th entry is the diagonal element $c_\mu^\mu = P_{\mu\mu}$. Expanding the reconstruction over the pattern basis,

$$\hat{f}^\mu = \sum_{\nu=1}^{N_{\text{patts}}} c_\nu^\mu f^\nu = c_\mu^\mu f^\mu + \sum_{\nu \neq \mu} c_\nu^\mu f^\nu,$$

so $(f^\mu)^\top \hat{f}^\mu = c_\mu^\mu (f^\mu)^\top f^\mu + \sum_{\nu \neq \mu} c_\nu^\mu (f^\mu)^\top f^\nu = c_\mu^\mu N_F + \sum_{\nu \neq \mu} c_\nu^\mu (f^\mu)^\top f^\nu$. Taking expectation over F (with H , hence c^μ , independent of F), every cross term $\mathbb{E}[(f^\mu)^\top f^\nu] = 0$ for $\nu \neq \mu$ by assumption (ii), leaving

$$\mathbb{E} \left[\frac{(f^\mu)^\top \hat{f}^\mu}{N_F} \right] = c_\mu^\mu = P_{\mu\mu}.$$

Averaging over all N_{patts} patterns, $\frac{1}{N_{\text{patts}}} \sum_{\mu} P_{\mu\mu} = \text{tr}(P)/N_{\text{patts}}$. P is an orthogonal projector, so its eigenvalues are all 0 or 1 and its trace equals its rank: $\text{tr}(P) = \text{rank}(P) = N_H$. Hence the average equals N_H/N_{patts} . $\square$

We verified this numerically too (random Gaussian H , i.i.d. $\{-1, 1\}$ columns for F) across four $(N_H, N_{\text{patts}}, N_F)$ triples spanning an order of magnitude in each dimension; measured and predicted overlap agreed to at least three decimal places in every case (e.g. $N_H=150$, $N_{\text{patts}}=1000$: measured 0.1500 vs. predicted 0.1500).

Discussion. Both theorems are corollaries of Lemma 3 applied to different matrices. Theorem 5.1 needs F full column rank ($N_{\text{patts}} \leq N_F$) and gets back case (a) – the hat matrix is the identity, so recovery is exact and deterministic, with no distributional assumption on either F or H needed at all. Theorem 5.3 needs H full row rank ($N_{\text{patts}} > N_H$, guaranteed by the scaffold's own construction) and gets back case (b) – the hat matrix is a rank-deficient projector, so recovery is only partial *on average*, and only under an added independence assumption on F (needed here specifically to make the projector's off-diagonal action on unrelated patterns cancel in expectation, since $P \neq I$ leaves no exact identity to fall back on). This is also exactly why Theorem 5.3, unlike Theorem 5.1, places no full-rank requirement on the *content* matrix F : the ceiling is set entirely by H 's rank relative to N_{patts} . Between the two theorems, MESH's own heteroassociation theory therefore covers both directions in which a pseudoinverse-learned heteroassociative map can be starved of rank: too few scaffold dimensions relative to the number of patterns (Theorem 5.3), or too little independent information in the content itself (Theorem 5.1, violated by real images in Section 4.5 of the main paper).

12.3 Why the same linear algebra looks catastrophic in Hopfield but graceful in MESH and our own architecture

Section 4.6 of the main paper reports that pseudo-inverse (projection-rule) Hopfield is “algebraically exact at zero noise” yet has basins of attraction that “shrink so fast under even modest noise” that recall crashes from 100% to 0% within a narrow range of M . We now show this is not a separate empirical curiosity but a direct, provable consequence of the same pseudo-inverse machinery above, applied to *auto*-association instead of heteroassociation – and that it is this specific consequence, not the pseudo-inverse rule in the abstract, that MESH’s scaffold and our own architecture avoid.

Setup. The classical projection-rule (pseudo-inverse) Hopfield weight matrix is

$$W = X(X^\top X)^+ X^\top \in \mathbb{R}^{N \times N},$$

the orthogonal projector onto the column space of the stored-pattern matrix $X \in \mathbb{R}^{N \times M}$ (columns x^μ).

Proposition 5 (Clean recall is always exact; noise correction is not). *For every stored pattern, $Wx^\mu = x^\mu$ exactly, at any M . For a noisy cue $\tilde{x} = x^\mu + n$, the reconstructed state is $W\tilde{x} = x^\mu + Wn$, and the expected fraction of noise energy removed by this single linear pass, for isotropic random noise n , is*

$$\frac{\mathbb{E}[\|(I - W)n\|^2]}{\|n\|^2} = \frac{N - \text{rank}(X)}{N} = \max\left(0, 1 - \frac{M}{N}\right),$$

which reaches exactly zero once $M \geq N$ and remains exactly zero for every $M > N$ thereafter.

Proof. Since $x^\mu = Xe_\mu \in \text{col}(X)$ trivially, and W is by construction the orthogonal projector onto $\text{col}(X)$, $Wx^\mu = x^\mu$ exactly – clean recall never degrades with M , unlike heteroassociation’s averaged overlap (Propositions 3–4). For the noise term: W and $I - W$ are complementary orthogonal projectors ($W(I - W) = 0$), of ranks $\text{rank}(X)$ and $N - \text{rank}(X)$ respectively, so by the Pythagorean identity $\|n\|^2 = \|Wn\|^2 + \|(I - W)n\|^2$ for every n . For an isotropic random unit vector n , the expected squared norm of its projection onto a fixed rank- r subspace is r/N (by symmetry, each of N orthogonal directions captures expected energy $1/N$); applying this to $I - W$ (rank $N - \text{rank}(X)$) gives $\mathbb{E}[\|(I - W)n\|^2] = (N - \text{rank}(X))/N$. Generic random patterns give $\text{rank}(X) = \min(M, N)$, so this is $\max(0, 1 - M/N)$, and since rank cannot exceed N , it is *exactly* zero, not merely small, for every $M \geq N$. $\square$

We verified Proposition 5 numerically (i.i.d. $\{-1, 1\}$ patterns, $N=500$, isotropic Gaussian noise): removed-noise fraction 0.8006 ($M=100$, predicted 0.8000), 0.3884 ($M=300$, predicted 0.4000), and exactly 0.0000 for every $M \in \{500, 600, 1000\}$ tested, against a predicted floor of exactly 0 once $M \geq N$.

The mathematical root of the asymmetry. Auto-association’s noise-correction capacity and heteroassociation’s reconstruction overlap are governed by *different* projector quantities. Auto-association’s noise correction depends on $\text{rank}(I - W) = N - \text{rank}(X)$ – an integer that hits zero exactly at a specific, finite M and cannot go lower, because rank is bounded below by zero. Heteroassociation’s overlap (Propositions 3–4) depends instead on the *average diagonal entry* of a projector, $\text{tr}(P)/N_{\text{patts}} = N_H/N_{\text{patts}}$ – a ratio that shrinks smoothly toward, but never exactly reaches, zero at any finite N_{patts} , and is provably monotone and convex in the load variable (Theorem 1, proved for our own architecture’s actual capacity-bearing readout; MESH’s Theorem 4.3 establishes the analogous robustness for its own scaffold, by a different, combinatorial argument). One is a hard, deterministic floor reached at a specific point; the other is a ratio that only approaches its limit.

Where the catastrophe actually comes from. Proposition 5 alone would predict only a hard-but-finite wall: at $M = N$ a single linear pass stops helping with noise at all, but nothing forces it to get *worse* beyond that point, nor explains sub-100% recall already at $M < N$ under noise. That amplification comes from the auto-associative network’s recurrent, nonlinear dynamics ($x \rightarrow \text{sign}(Wx)$, iterated to a fixed point), whose basin-of-attraction structure near this same $M \approx N$ boundary is known, via the classical statistical mechanics of Hopfield networks, to undergo its own sharp transition – the Amit–Gutfreund–Sompolinsky storage-capacity analysis and Kanter & Sompolinsky’s projection-rule result [8, 9], already cited in Section 4.6 of the main paper for exactly this reason. We do not re-derive that transition here; what Proposition 5 establishes rigorously is the boundary condition feeding into it: the linear noise-correction mechanism is provably, exactly exhausted at $M = N$ (or, for real correlated content such as the MNIST images of Section 4.5, at the content’s own effective rank rather than its nominal dimension) – precisely where that separately-established recurrent instability becomes catastrophic rather than merely diminished.

Why MESH and our own architecture do not inherit this wall. Neither routes an imperfect reconstruction back into a same-representation recurrent map built by projecting stored patterns onto their own linear span. MESH’s scaffold dynamics are a k -winners-take-all attractor network over a fixed random projection, with basins proven maximally large and stable for *every* N_{patts} up to $\binom{N_L}{k}$ (MESH’s Theorem 4.3) regardless of how imperfect the heteroassociative step feeding it is. Our own core engram and attention readout (Sections 2–3 of the main paper) involve no pseudo-inverse-learned recurrent map at all – the engram is a fixed random projection plus k -winners-take-all, and its capacity-bearing quantity (Section 2.3, and Theorem 1 above) is provably smooth and convex with no finite-size singularity. Both avoid Proposition 5’s wall not by a better learning rule, but by never subjecting the capacity-determining step to it in the first place.

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